

Scalability & Future Plan

Date	03 October 2026
Team ID	17
Project Name	Fitbuddy-AI Fitness plan Generator
Department	Computer Technology
Institution / University	Madras Institute Of Technology / Anna University
Academic Year / Semester / Section	3 / 5 / Batch A
Faculty Mentor / Guide	Jayachitra V P
Team Leader	Bharath Jeyakkumar S
Sprint 1 Start / End	28-09-2026 / 28-09-2026
Sprint 2 Start / End	29-09-2026 / 30-09-2026
Sprint 3 Start / End	01-10-2026 / 02-10-2026
Team Roles	Abdul Wahith M — Backend & AI Integration; Kamalesh T — UI / Frontend Development; Bilu Besto H — Validation, Feedback & Reliability; Bharath Jeyakkumar S — Team Leader / Testing & Quality Assurance; Prasannavelan K — Database, Documentation & Demo Support
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	SQLite local database	Limited concurrent write scalability	High
2	Gemini generation is request-latency sensitive	End-to-end latency depends on external AI service	High
3	Local admin view has no authentication layer	Not suitable as a production admin console	High
4	Fitness profile is relatively small	Less personalization than a full health/fitness profile	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local FastAPI + SQLite	Multiple workers behind a reverse proxy
2	Data Storage	SQLite file	PostgreSQL with connection pooling
3	AI Requests	Synchronous generation	Async client/ background jobs and queues
4	Security	API key in environment; local admin view	Authentication, authorization, rate limiting and secrets manager
5	Observability	Basic logging	Structured logs, metrics and tracing

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Exercise library, progress tracking and plan history versions	1-2 months	Better continuity
Phase 3	PostgreSQL, authentication and hosted deployment	2-3 months	Multi-user production readiness
Phase 4	Wearable integration, charts, reminders and nutrition tracking	3-6 months	Broader fitness use cases